

A short summary of how we can approach the modelling of rooms and a house

Goal: we model each room as a thermal network form resistances and capacitances, similar to an electric circuit.

Example of a room:

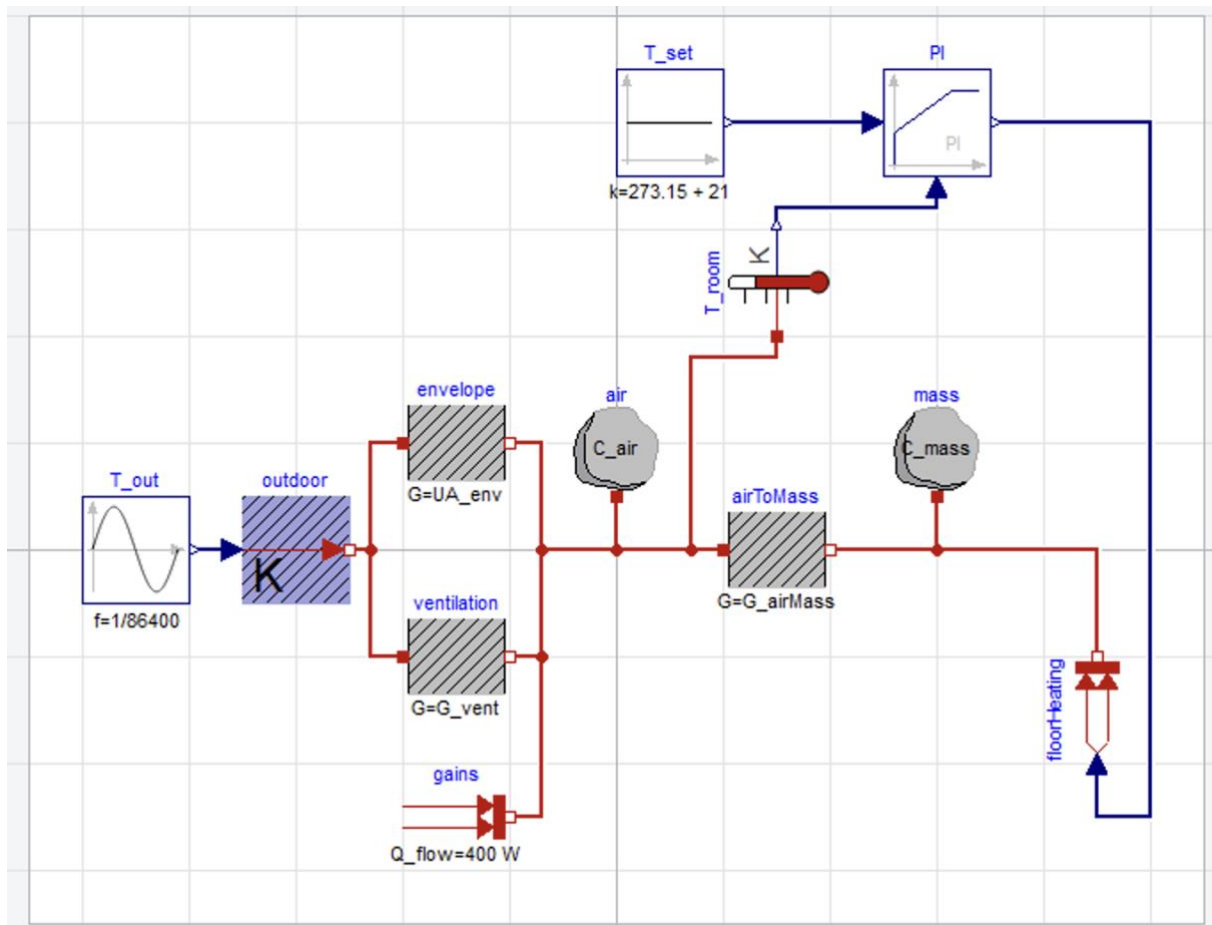


Figure 1: made by AI. Variables are visible in code view. (Bottom left button)

- Two Capacitances C_{air} , C_{mass} (walls and floor) – Those basically store heat
- Outdoor temperature on the left is later our weather data, here just a sin input that is converted to a heat signal. The blue “outdoor” is the translation of signals
- Envelope and Ventilation are both heat losses to the outside.
 - o Envelope: Heat through walls, windows, roof, floor
 - o Ventilation: warm air exchanged with cold air
 - o Param G [W/K] is the constant thermal conductance of the material
- Room air and mass exchange heat by convection (“AirToMass”)
- Internal gains (bottom) heat the air (appliances, people, ...) , floor heating heats the mass
- PI Controller adjusts the floor heating power to keep it at 21°C

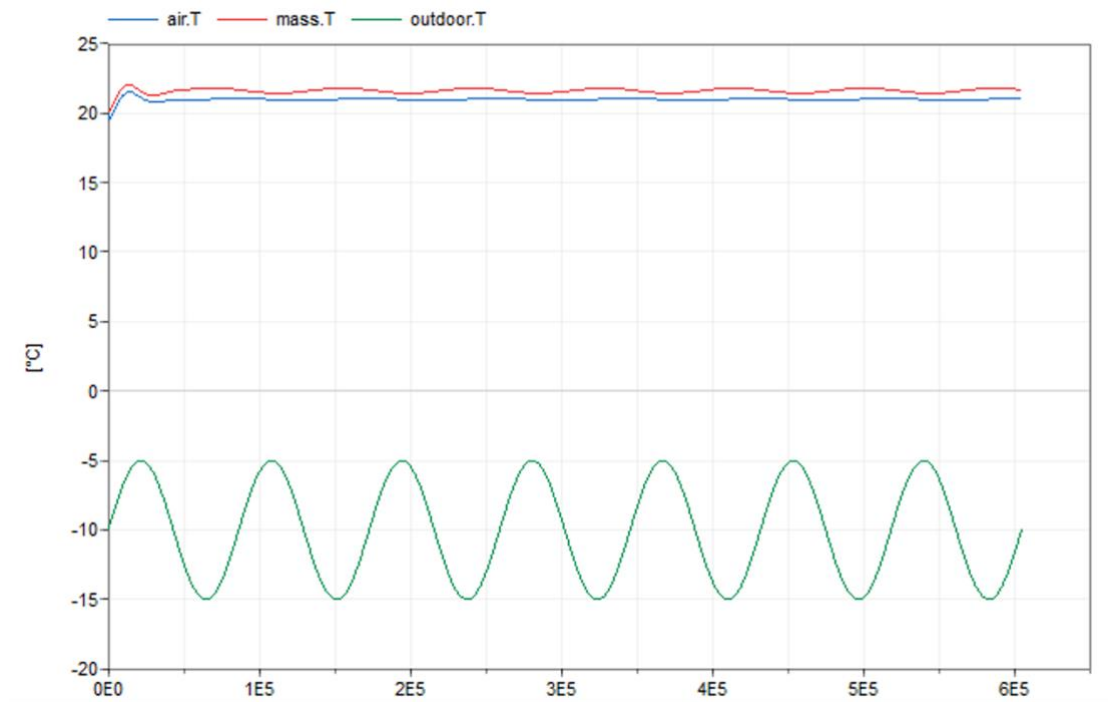


Figure 2: Temperature of air (blue) and mass (walls, floor, ...) in the room and outside air temperature over one week

Components of the system:

Component	Electrical	In a House	Dymola	Usage
Resistor (R)	Resistor	Wall, Window, Air exchange, Ventilation system	<i>Thermal Conductor</i>	Slows the heat flow down
Capacitance (C)	Capacitor	Room Air, Walls, Furniture	<i>HeatCapacitor</i>	Can absorb and give off heat
Temp Source	Volt Source	Outdoor air, ground (from weather input)	<i>Prescribed-Temperature</i>	Fixed Temperature. Like the constant values/ramps we had
HF Source	Current Source	Heating, occupants, appliances, solar gains/absorption	<i>PrescribedHeatFlow</i>	Pushes watts into a node
Heat pump / Storage	Controlled current source	Heat generation, domestic hot water	<i>TIL components</i>	Modelled like in the assignments. Complex inside and delivers heat into a node
Node	Node	Point with one temperature (T _{air} for example)	<i>HeatPort</i>	Energy balance: in – out = stored

How we can approach the modelling of the house:

- Each room is its own model and must work on their own first.
- Each room integrates the component from the picture:
 - room air (C),
 - thermal mass (C),
 - outer envelope (R to outdoor temperature),
 - ventilation (R),
 - windows (R + solar gain),
 - floor heating (heat source into the floor mass),
 - internal gains (heat source into the air)
- Walls to adjacent rooms: R-C-R. Also Convection, Conduction, Convection.
 - That is summarized by the “U” value in the task given. We can just use that, it combines all three steps into one.
- Open doors: a simple R between the air nodes of two rooms
- Test each room alone with fixed boundary conditions, then connect them together
- The special cases are added later (modelled for example with the TIL library) and connected to the rooms through the same heat ports.
 - So a heat pump cycle can be integrated similarly to assignment 4

In short: sources drive temperature, resistances slow down and capacitances delay

Open Questions:

If you search online, often times the “buildings” library by modelica itself is used. I did not look much into it. I will ask the TA next exercise if we should use it or not.